TOWARDS EXPLAINABLE AI: INTERPRETABLE MODELS AND FEATURE ATTRIBUTION

Tapomoy Adhikari

tapo3930@gmail.com
tapomoy_adhikari@outlook.com

Abstract

In recent years, the development of Explainable Artificial Intelligence (XAI) has been deemed an important area of research, driven by the need to build trust and transparency in AI systems. The goal of XAI is to enable the understanding of the decision-making process of complex AI models by humans. An overview of two main approaches to XAI: interpretable models and feature attribution, is presented in this paper. Interpretable models aim to build models that are inherently interpretable, while feature attribution methods aim to identify which features or inputs are most important in the decision-making process. Despite significant progress made in XAI, there are still several challenges that need to be addressed, including the development of more accurate and interpretable models, more efficient and scalable methods for feature attribution, and ensuring that XAI is used responsibly and ethically.

Introduction

Artificial Intelligence (AI) has revolutionized many domains, from natural language processing to image and speech recognition. However, as AI is increasingly used in critical domains such as healthcare, finance, and autonomous vehicles, there is a growing need for transparency and interpretability. Interpretability refers to the ability to explain the decisions made by AI models in a way that is understandable to humans. It is important because it can help build trust in AI systems and improve their adoption.

There are two main approaches to interpretable AI: interpretable models and feature attribution. Interpretable models are models that are inherently interpretable, meaning that their structure and parameters can be easily understood and interpreted by humans. Feature attribution, on the other hand, refers to the methods used to identify the features or inputs that are most important in the decision-making process of a black box model.

Interpretable Models

Interpretable models are models that are designed to be transparent and interpretable by humans. These models are often referred to as white box models because their internal workings can be easily understood. Some examples of interpretable models include decision trees, linear models, and rule-based models.

Decision trees are a type of model that can be easily visualized and interpreted. They consist of a series of decision nodes that split the data based on different features. The leaves of the tree represent the final decision. Decision trees are commonly used in domains such as healthcare and finance, where interpretability is important.

Linear models are another type of interpretable model that are widely used in machine learning. They consist of a linear combination of features that are weighted by coefficients. The coefficients represent the importance of each feature in the decision-making process. Linear models are simple to interpret and can be used for both classification and regression tasks.

Rule-based models are models that consist of a set of rules that are used to make decisions. The rules are typically expressed in the form of “if-then” statements. For example, a rule-based model in healthcare might say “if a patient has high blood pressure and diabetes, then prescribe medication

X". Rule-based models can be easily understood by humans and can be used in domains where transparency and interpretability are important.

Interpretable models have several advantages over black box models. They are transparent, which means that their internal workings can be easily understood and interpreted. This can help build trust in the model and improve its adoption. Interpretable models are also often simpler than black box models, which means that they can be trained with less data and computational resources.

However, interpretable models also have some limitations. They are often less accurate than black box models, which means that they may not perform as well on complex tasks. Interpretable models can also be limited by their ability to capture complex relationships between features.

Feature Attribution

Feature attribution refers to the methods used to identify the features or inputs that are most important in the decision-making process of a black box model. Black box models are models that are difficult to interpret because their internal workings are not easily understood. Examples of black box models include deep neural networks and support vector machines.

Feature attribution methods aim to provide insights into how a black box model is making decisions. One approach to feature attribution is the use of gradient-based methods, such as Integrated Gradients and DeepLIFT. These methods calculate the gradients of the output with respect to the inputs and use these gradients to identify which inputs are most important in the decision-making process.

Another approach to feature attribution is the use of model-agnostic methods, such as LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (Shapley Additive Explanations). These methods generate local explanations for individual predictions by training interpretable models on subsets of the data. The interpretable models are then used to explain the predictions of the black box model.

Feature attribution methods have several advantages. They can provide insights into how a black box model is making decisions, which can help build trust in the model and improve its adoption. Feature attribution methods can also be used to identify biases in the data or model and to improve fairness.

However, feature attribution methods also have some limitations. They can be computationally expensive and may not work well on all types of models. Feature attribution methods can also be limited by their ability to capture the complexity of the decision-making process.

Challenges and Future Directions

While interpretable AI has made significant progress in recent years, there are still several challenges that need to be addressed to make AI more transparent and interpretable. One of the main challenges is to develop more accurate and interpretable models that can perform well on complex tasks. Another challenge is to develop more efficient and scalable methods for feature attribution.

Another challenge is to ensure that interpretable AI is used responsibly and ethically. Interpretable AI can be used to identify biases in the data or model and to improve fairness, but it can also be used to perpetuate biases if not used properly. It is important to ensure that interpretable AI is used to promote transparency, accountability, and fairness.

Conclusion

Interpretable AI is an important area of research that can help build trust in AI systems and improve their adoption. Interpretable models and feature attribution are two main approaches to interpretable AI, each with its own advantages and limitations. There are still several challenges that need to be addressed to make AI more transparent and interpretable, but with continued research and development, interpretable AI has the potential to revolutionize many domains.